Project - 1: Railway Guages Data Analysis

This project demonstrates how to impact CSV data into Pandas, analyze it, and visualize results using Matplotlib.

Pandas Overview:

Pandas is a powerful data analysis library used for data manipulation and analysis.It provides DataFrame structures for handling tabular data.Pandas is widely used for data analysis because it provides simple and flexible methods to handle structured data.

First,we need to import the required libraries and load the dataset.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# load CSV file
df = pd.read_csv('railway_gauges.csv')

print(df.head())
```

After this,Pandas converts the data into a **DataFrame**,which makes it easy to view and manipulate data.Using **head()** we can see the first 5 rows,and similarly **tail()** can be used to view last 5 rows.

Understanding the Data

The dataset contains details like:

i. Year
ii. Different types of gauges(Broad,Metre,Narrow)
iii. Total number of tracks

Example representation:

	Year	Broad Gaug	Metre Gaug	Narrow Gai	Total
2	1964-65	3521	2891	464	6876
3	1965-66	3603	2913	470	6986
4	1966-67	3615	2911	481	7007
5	1967-68	3601	2920	506	7027

Useful Pandas Operations:

Pandas provides different ways to access data:

- loc[] is used for accessing row using index

- iloc[] is used for accessing specific column

To find important insights,we can use built-in functions.For exampleto get the year with maximum installations.

```
# find which year had the max installations
print(df.iloc[[df['Total'].idxmax()]])
```

This will return the row where the total number of tracks is highest.

Data Visualization:

To better understand the data,we can represent it graphically.Here,a bar chart is used to show how railway tracks were installed over the years.

```
# Plot data using bar chart
df = df.drop('Total', axis=1)
ax = df.plot(x='Year', kind="bar")

plt.xticks(rotation=70)
plt.xlabel('Year')
plt.ylabel('Number of Tracks')
plt.title('Gauges: Number of railway tracks installed per year')

plt.savefig('rail_gauges.png')
plt.show()
```

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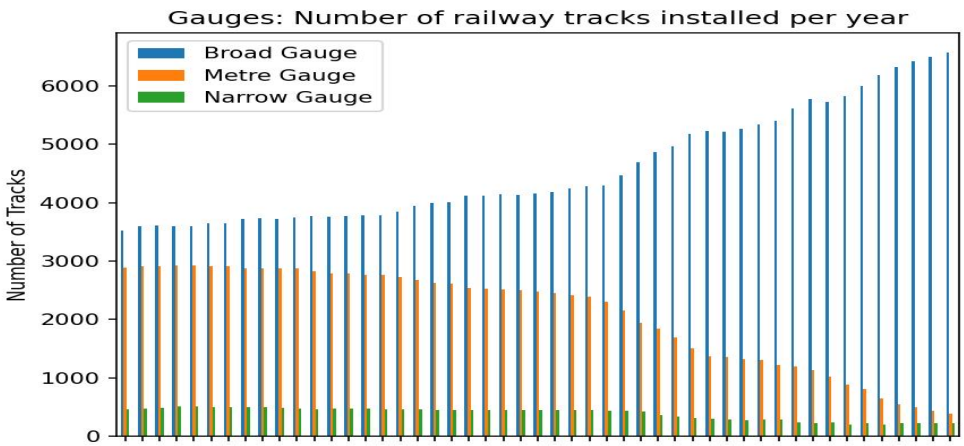
The graph clearly shows the variation in the track installation across different years and gauge types.

Result:

- ✓ Identified the year with highest railway track installation.
- ✓ Visualized the data using a bar chart
- ✓ Understood how Pandas simplifies data analysis.

And the output looks like below.



Conclusion:
Through this project ,we learned how to:
1. Import and read CSV filies
2. Perform basic data analysis
3. Extract useful insights
4. Represent data visually

This project helped me understand the data clearly and interpret the trends effectively using visualization.